

JOHN LEWIS

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EDUCATION

University of Virginia, Charlottesville, VA

August 2024 - May 2028

B.S. Mechanical Engineering, Minor in Electrical Engineering

GPA: 3.76/4.00

- Relevant Coursework: Aerospace Structures, Fluid Mechanics, Thermal Fluids Lab, Thermodynamics, Signals and Systems, Applied Circuits, Strengths of Materials, Dynamics, Mechanics Lab, Computer-Aided Design
- Activities: Virginia Alpine Ski and Snowboard Team, UVA Triathlon Club, UVA Surfrider, UVA Cycling

RELEVANT EXPERIENCE

Undergraduate Researcher, **Center for Applied Biomechanics**, Charlottesville, VA

May 2026 - Present

- Designed multiple motorcycle helmet impact test rigs in SolidWorks, including an inverted pendulum simulating rider rotation at speeds up to 12.2 m/s and two drop-tower systems to compare international helmet standards
- Produced machine-shop drawings and performed FEA to validate components under impact loading
- Designed and 3D printed a shoulder surrogate integrated with a Hybrid III 50th-percentile male head form to investigate shoulder effects on helmeted head kinematics during motorcycle impacts
- Conducted instrumented helmet testing using VICON motion capture, Hybrid III sensors, load cells, and DAS, measuring linear and rotational acceleration and velocity during impact
- Processed and visualized experimental data in MATLAB, compared head-response metrics across test configurations, and presented findings to faculty

Embedded Systems Project, **Egg Theft Detection System**, Charlottesville, VA

May 2026

- Designed and built an ESP32-based surveillance system using a thin-film force sensor to detect egg-carton removal and automatically trigger image capture
- Implemented ESP-NOW communication between an ESP32 and ESP32-CAM and developed a Flask web server to receive, store, and display captured images over Wi-Fi
- Designed and 3D printed custom enclosures and sensor mounts, soldered the final circuit onto perfboard, and integrated battery-powered voltage regulation for standalone operation

Team Lead, **Potential Energy Propulsion Racer**, Charlottesville, VA

October 2025 - December 2025

- Led the design and prototyping of a 3D-printed, spring-powered obstacle-course vehicle constrained to a $12 \times 8 \times 4$ in. starting envelope
- Designed a 12:1 compound gearbox, guided spring slider, bearing-supported axles, and deployable scoring arms in SolidWorks
- Iteratively tested and refined drivetrain, traction, and obstacle-clearing performance, helping the vehicle score 1,200 points and place 3rd overall

Mechanical Lead, **Gizmologists: University of Virginia**, Charlottesville, VA

August 2024 - Present

- Lead mechanical design and CAD organization for a 15-member multidisciplinary team focusing on small mechatronic projects
- Designed and built a clock with a gantry, stepper motors, and servo motor with a stylus to write time on pressure-sensitive boards using an Arduino microcontroller and C/C++
- Train new members in SolidWorks, prototyping, and mechanical design practices to increase participation in club projects

Chassis Team Member, **Virginia Motorsports**, Charlottesville, VA

August 2024 - May 2026

- Modeled chassis components in Fusion 360 according to Formula SAE packaging, structural, and competition requirements
- Performed FEA to evaluate chassis stress, deformation, and factors of safety under defined loading conditions
- Manufactured chassis components using CNC mills, waterjet cutting, and TIG welding

WORK EXPERIENCE

Server and Bartender - Seasonal, **Shelter Island Yacht Club**, Shelter Island, NY

June 2024 - August 2026

- Delivered high-volume food and beverage service while coordinating with a 30-person hospitality staff
- Maintained detailed knowledge of menu offerings and cocktail preparation while responding efficiently to member requests

SKILLS & CERTIFICATIONS

- Technical: SolidWorks, Fusion 360, MS Office, Autodesk CFD, ANSYS Mechanical
- Programming Languages: MATLAB, C++, Python, Java, C#
- Certifications: Certified SolidWorks Associate obtained through *Dassault Systems* in October 2025